

$$N = p_1^{e_1} \dots p_k^{e_k}$$

$$f(N) = f(p_1^{e_1}) \times f(p_2^{e_2} \times \dots \times p_k^{e_k})$$

$$\quad \quad \quad \parallel$$

$$f(p_2^{e_2}) \times f(p_3^{e_3} \times \dots \times p_k^{e_k})$$

$$\vdots$$

for example:

~~完全~~ 完全积性

$$f(n) = \prod_{k=1}^K c_k \quad (c_k - \text{constant}) \quad \text{完全积性}$$

$$\text{令 } n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k} \\ (p_1 < p_2 < \dots < p_k \\ e_1, e_2, \dots, e_k \geq 1)$$

$$f(n) = p_1 p_2 \dots p_k$$

积性

$$g(n, m) = 1$$

存在质数 p $\Updownarrow$ 同时 $p|n$ 且 $p|m$

$1 \leq d \leq n$, $\lfloor \frac{n}{d} \rfloor$ 的取值

$$\textcircled{1} \quad 1 \leq d \leq \sqrt{n} \quad \lfloor \frac{n}{1} \rfloor \quad \lfloor \frac{n}{2} \rfloor \quad \dots \quad \lfloor \frac{n}{m} \rfloor$$

$$m = \max_{m \leq \sqrt{n}} m^2 \leq n \quad \rightarrow O(\sqrt{n})$$

$$\textcircled{2} \quad d > \sqrt{n} \quad \frac{n}{d} \leq \sqrt{n} \Rightarrow 1 \leq \frac{n}{d} \leq \sqrt{n} \rightarrow O(\sqrt{n})$$

$$\lfloor \frac{n}{d} \rfloor \rightarrow O(\sqrt{n}) \text{ 种}$$

$$\phi(n) = \sum_{i=1}^n [\gcd(n, i) = 1]$$

1~n 中与 n 互质的个数

$$[P] = \begin{cases} 1 & P \text{ is true} \\ 0 & P \text{ is false} \end{cases}$$

$\phi(n)$ 是一个积性函数

积性

$$n = p_1^{e_1} \times p_2^{e_2} \times \dots \times p_k^{e_k}$$

$$f(n) = f(p_1^{e_1}) \times f(p_2^{e_2}) \times \dots \times f(p_k^{e_k})$$

$$\phi(n) \rightarrow \phi(p^e) = 1 - p^e \text{ 里不是倍数的个数} = \text{总数} - \text{是倍数的}$$

$$= p^e - p^e \times \frac{1}{p} = p^{e-1} \times (p-1)$$

$$\gcd(n, m) = d \Leftrightarrow \gcd\left(\frac{n}{d}, \frac{m}{d}\right) = 1$$

$$\sum_{d|n} \phi(d) = \sum_{d|n} \phi\left(\frac{n}{d}\right) = \sum_{d|n} \sum_{\substack{i=1 \\ \frac{n}{d} \mid i}} \left[\gcd\left(\frac{n}{d}, i\right) = 1 \right]$$

$$n=10 \quad \begin{array}{l} d=1, 2, 5, 10 \\ \frac{n}{d}=10, 5, 2, 1 \end{array}$$

$$= \sum_{d|n} \sum_{\substack{i=1 \\ \frac{n}{d} \mid i}} \left[\gcd(n, id) = d \right] = n$$

$$f \xrightarrow{\text{积性}} \sum_{d|n} f(d) \xrightarrow{\text{积性}} \sum_{d|n} \phi(d) \xrightarrow{\text{积性}}$$

$$g(p^e) = \phi(1) + \phi(p) + \phi(p^2) + \dots + \phi(p^e) \\ = 1 + (p-1) + (p-1)p + \dots + (p-1)p^{e-1} \\ = p^e$$

$$g(n) = g(p_1^{e_1}) \times \dots \times g(p_k^{e_k}) = p_1^{e_1} \times \dots \times p_k^{e_k} = n$$

$$E(n) = \begin{cases} 1 & n=1 \\ 0 & n>1 \end{cases}$$

$$\text{Id}_K(n) = n^*$$

$$\underline{I}(n) = 1$$

$$\phi * \underline{I} = \underline{Id}$$

$$I(n) = 1$$

$$Id(n) = n$$

$$I * f(n) = \sum_{d|n} f(d)$$

$$I * 1 = Id$$

```
1 for (int s = 1, t; s <= N;)
2 {
3     t = N / (N / s);
4     //N / s, N / (s + 1), ..., N / t相等
5     //i = s ~ t, N / i = N / s
6     Ans += f(N / s) * (t - s + 1);
7     s = t + 1;
8 }
9 //数论分块
10 //复杂度是 $O(\text{计算}f) * O(N^{0.5})$ 
```

$$h(n) = \sum_{a \cdot b = n} f(a) \times g(b)$$

$$h(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right)$$

$$f(n) = \begin{cases} 1 & n=1 \\ 0 & n>1 \end{cases} \quad f * f(n) = \sum_{d|n} f(d) \times f\left(\frac{n}{d}\right) = f(n)$$

当 a 与 p 互质, $a^{\phi(p)} \equiv 1 \pmod{p}$

$$a^n \equiv a^{n \bmod \phi(p)} \pmod{p}$$

特别地, p is a prime, $\phi(p) = p-1$

更一般地, a 与 p 不一定互质

$$a^n \bmod p = \begin{cases} a^n & n < \phi(p) \\ a^{n \bmod \phi(p)} \times a^{\phi(p)} & n \geq \phi(p) \end{cases}$$

Dirichlet卷积

```
//h(1), h(2), ..., h(n) //h = f * g for (int a = 1; a <= n; ++a) for (int b = 1; a * b <= n; ++b) h[a * b]
+= f[a] * g[b]; //h(1), h(2), ...,
h(n) //h = f * g for (int a = 1; a <= n; ++a) for (int b = 1; a * b <= n; ++b) //O(n / a) h[a * b] +=
f[a] * g[b]; //O(n / 1) + O(n / 2) + ... = O(n ln(n))
```

不定方程解数计数题解

$$1 \leq x_1 \leq x_2 \leq \dots \leq x_n = M$$

满足, 任意 i , 有 $x_{i+2} - x_i \leq 1$

$$0 \leq d_i \leq 1 \quad d_i = x_i - x_{i-1}$$

$$x_{i+2} - x_i = d_i + d_{i+1} \leq 1$$

$$d_i \in \{0, 1\}$$

有恰好 M 个 d_i 为 1

同时, 相邻两个 1 间距超过 0

x_i 为第 i 个 1 和第 $i-1$ 个 1 之间的距离

$$\underbrace{y_1}_{y_1 \geq 0} + \underbrace{x_2}_{x_i \geq 1} + \underbrace{x_3}_{x_i \geq 1} + \dots + \underbrace{x_M}_{x_i \geq 1} + \underbrace{y_{M+1}}_{y_{M+1} \geq 0} = \text{len} - M$$

不定方程解计数

$$x_0 = 1$$

$$d_i = x_i - x_{i-1}$$

$$d_1 + d_2 + \dots + d_n = M$$

$$x_1 - x_0 + x_2 - x_1 + \dots + x_n - x_{n-1} = x_n = M$$

||
|

二项式定理

来看一道例题：令 $x = 1 - y$ ，同时 $\sum_{i=0}^n a_i \times (2 - y)^i = (x + 3)^n$
请你求出 a_k

$$t = 2 - y \quad x + 3 = 4 - y = t + 2$$

$$\sum_{i=0}^n a_i \times \binom{n}{i} t^i = (t + 2)^n$$

拆分数

```
for (i) for (j) { F[i][j] = F[i - j][j - 1] + F[i - j][j]; if (i >= K) F[i][j] -= F[i - K][j - 1]; }
```

求排列数和组合数具体值的简单方法

```
//Fac[n] = n! //Inv[n] = 1 / n! //brute force /* inverse(a) returns the inverse of a in other
words, inverse(a) * a = 1(mod P) inverse(a) ---- O(log_2(P)) */ Fac[0] = 1; for (int i = 1; i <=
N; ++i) { Fac[i] = Fac[i - 1] * i; Inv[i] = inverse(Fac[i]); } //O(N * log_2(P))//advanced Fac[0] = 1;
for (int i = 1; i <= N; ++i) Fac[i] = Fac[i - 1] * i; Inv[N] = inverse(Fac[N]); for (int i = N - 1; i >=
0; --i) Inv[i] = Inv[i + 1] * (i + 1); //1 / i! = (i + 1) / (i + 1)! //O(N + log_2(P))
```

$$\underbrace{(Id * \phi)(n)}_{h(p^e)} \quad , \quad \underbrace{h}_{\text{积}} = \underbrace{Id}_{\text{积}} * \underbrace{\phi}_{\text{积}} \quad h(n)$$

$$h(p^e) \rightarrow h(n) = h(p_1^{e_1}) h(p_2^{e_2}) \dots h(p_k^{e_k})$$

$$\begin{aligned}
& \sum_{i=1}^n \gcd(n, i) \\
&= \sum_{d|n} d \times \sum_{i=1}^n [\gcd(n, i) = d] \\
&= \sum_{d|n} d \times \sum_{i=1}^{\frac{n}{d}} [\gcd(\frac{n}{d}, i) = 1] \quad = \sum_{d|n} d \phi\left(\frac{n}{d}\right)
\end{aligned}$$

积性函数

f 积性函数 数论函数

$$\forall a, b \in \mathbb{N}^+ \text{ 且 } \gcd(a, b) = 1$$

$$\text{有 } f(a) \times f(b) = f(ab)$$

$$(a, b \in \mathbb{N})$$

完全积性函数

$$\forall a, b \in \mathbb{N}^+$$

$$\text{有 } f(a) \times f(b) = f(ab)$$

(属于积性函数)

组合意义

$$a^n$$

$a, n \geq 0$

组合

意义

n — steps

step — a ways

$$\underbrace{a \times a \times \dots \times a}_{n\uparrow} = a^n$$